Self-Supervised Temporal Super-Resolution Framework With METEOR-Net for Enhancing Convective Process Reconstruction in Geostationary Satellite Cloud Data

Xiang Lin, Yunying Li, Zitong Chen, and Xiongwei Kou

***Abstract*—Next-generation geostationary satellite FY-4A provides cloud data with a temporal resolution of 15 min, while shallow convection may evolve into deep convection in less than 30 min. This temporal sparsity limits the satellite's ability to capture rapid convective processes, thereby hindering effective weather monitoring and early warning. Temporal super-resolution technology is used to address this limitation, but existing temporal super-resolution methods often directly apply video frame interpolation (VFI) paradigms designed for RGB images, resulting in transfer errors. To address such issues, a novel spatiotemporal self-supervised temporal super-resolution framework is proposed, which leverages the spatiotemporal consistency between the regional and full-disk observations of the FY-4A satellite to establish a temporal super-resolution mapping relationship without the need for manually labeled data. In addition, a multichannel encoding and temporal-flow evolution network for optimized-observation reconstruction (METEOR-Net) is proposed, which addresses the challenges of multichannel asynchrony and complex nonlinear evolution in geostationary satellite cloud data through multichannel collaborative encoding (MCE), temporal-flow estimation and evolution (TFEE), and evolution-aware feature fusion mechanisms. To enhance the focus on convective regions, a new convective region-weighted loss function and evaluation metrics are designed. Experimental results demonstrate that this method achieves state-of-the-art (SOTA) performance in visual quality and convective process reconstruction accuracy in the FY-4A satellite cloud data temporal super-resolution reconstruction task, successfully enhancing the temporal resolution of FY-4A cloud data from 15 to 5 min. This provides high-resolution satellite cloud data for future convective weather monitoring and early warning systems.**

***Index Terms*—Multichannel collaborative, self-supervised learning, temporal super-resolution, temporal-flow evolution.**

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I. Introduction

METEOROLOGICAL satellites are deployed in space to monitor Earth's atmosphere, oceans, and surface conditions. These satellites are equipped with advanced remote sensing instruments that capture reflectance and radiation intensity at specified spectral wavelengths and spatiotemporal resolutions, providing essential information support for various meteorological operations. Based on their orbital characteristics, meteorological satellites are categorized into two types: polar-orbiting satellites [1], [2], [3], [4] and geostationary satellites [5], [6], [7]. Polar-orbiting satellites operate in low-Earth polar orbits, enabling comprehensive coverage of every region on Earth within a fixed time period. These satellites are particularly suitable for long-term climate monitoring and polar observations. However, their revisit time of approximately 12 h precludes continuous meteorological monitoring of specific regions. In contrast, geostationary satellites operate in geostationary orbit, allowing sustained observation of the same region, making them ideal for real-time weather monitoring and early warning systems.

Next-generation geostationary satellites, including GOES-16 [5], Himawari-8 [6], and FY-4A [7], are capable of full-disk observations and providing cloud data. For example, FY-4A provides geostationary satellite cloud data with a temporal resolution of 15 min, based on its 15-min revisit time. While this temporal resolution is sufficient for studying long-term atmospheric trends, climate models, and other gradually evolving processes [8], [9], [10], [11], it proves inadequate for rapidly developing convective events, such as convective processes that may evolve from shallow convection to deep convection in just 30 min. When relying solely on full-disk data, only the initial and final states of different convective stages are captured, while the actual rapid evolution process within a revisit period remains unobserved, which limits the tracking and early warning for convective events. Consequently, convective weather monitoring and warning research is limited to mesoscale systems where temporal high-resolution satellite observations are available [12], [13], [14], making large-scale implementation challenging. This limitation is referred to as the temporal sparsity dilemma of geostationary satellites. Solving this dilemma requires temporal high-resolution data capable of capturing the actual convective evolution process.

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However, due to technical and economic constraints, current geostationary satellites cannot acquire continuous full-disk data at high temporal resolutions. Therefore, temporal super-resolution technology, which enhances the temporal resolution of data through algorithms, offers a preferred solution without the need to modify the hardware.

Traditional geostationary satellite cloud data processing often uses linear interpolation methods [15] for temporal super-resolution. These methods generate intermediate frames by a weighted combination of adjacent geostationary satellite cloud data frames, thereby increasing the frame rate within the same time period. However, such methods do not extract incremental information from geostationary satellite cloud data, fail to model complex cloud motion dynamics effectively, making it challenging to obtain sufficiently reliable temporal high-resolution data for practical applications.

In the exploration of optimal temporal super-resolution solutions for geostationary satellite cloud data, artificial intelligence techniques based on spatiotemporal self-supervised learning have been increasingly applied to meteorological data mining and utilization [16], [17], [18], [19], [20], [21]. This self-supervised, data-driven paradigm, which does not rely on labeled data, offers new insights for geostationary satellite cloud data temporal super-resolution.

Due to the similarity between satellite temporal super-resolution and video frame interpolation (VFI), VFI methods based on spatiotemporal self-supervised learning have gained significant attention. VFI methods based on spatiotemporal self-supervised learning are primarily categorized into two types: kernel or phase-based direct generation methods without flow [22], [23], [24], [25], [26], [27], [28], [29], [30], and flow-based generation methods [31], [32], [33], [34], [35], [36], [37], [38], [39], [40]. Compared to kernel or phase-based direct generation methods, flow-based methods have become mainstream due to their ability to maintain motion consistency through flow estimation, thereby avoiding incorrect and incoherent spatial displacements. The key idea of flow-based generation methods lies in decomposing the VFI process into flow estimation and flow-guided frame reconstruction. Most studies adopt flow-based generation methods, where neural networks are employed to estimate optical flow between adjacent frames as explicit motion representations, followed by flow-guided warping for intermediate frame reconstruction. Dosovitskiy et al. [41] pioneered data-driven optical flow estimation using convolutional neural networks. Subsequently, Jiang et al. [31] proposed SuperSlomo, using a UNet architecture for bidirectional optical flow estimation and frame reconstruction through flow-based warping for multiframe interpolation. Later research largely adopted this optical flow-based interpolation paradigm, and focused on achieving more accurate flow estimation [32], [35], [37], [38], [40].

With the maturation of spatiotemporal self-supervised VFI technology, Vandal and Nemani [42] formulated the temporal super-resolution of geostationary satellite cloud data as a VFI task, adopted the VFI framework and optical flow-based modeling paradigm, adapting the SuperSlomo-T (SSM-T) model to geostationary satellite cloud data temporal super-resolution through parameter fine-tuning and retraining. Experimental results show that this method achieves higher accuracy compared to traditional super-resolution methods.

However, despite the proven feasibility of transferring spatiotemporal self-supervised VFI methods to geostationary satellite cloud data temporal super-resolution, direct application of VFI frameworks and optical flow-based modeling paradigms presents three significant challenges due to fundamental differences in data characteristics and task objectives as follows.

First, a data construction challenge. As a data-driven method, the learning capacity of deep learning models is bounded by the characteristics of training data. While video frames can be readily captured at varying frequencies using low-cost recording devices, enabling the acquisition of continuous high and low-frequency images for learning VFI mapping relationships, geostationary satellites, with their high operational costs, are limited to low-frequency full-disk observation at 15-min intervals. Ensuring the reliability of the super-resolution mapping relationship constructed from geostationary satellite cloud data under these constraints presents the primary technical challenge that must be addressed. Notably, previous studies have not conducted systematic analyses of such reliability considerations.

Second, model adaptation requirements. The spatiotemporal evolution characteristics of geostationary satellite cloud data differ significantly from video data, potentially introducing transfer errors when directly applying optical flow-based VFI paradigms. These differences manifest in two aspects. First, geostationary satellite cloud data exhibits multichannel temporal asynchrony. Unlike RGB video frames, whose channels exhibit synchronized spatiotemporal variations in primary change regions as shown in Fig. 1(a), geostationary satellite cloud data shows the asynchronous temporal evolution dynamics of different channels as illustrated in Fig. 1(b). These channels form complementary spatiotemporal relationships, exemplified by mid-level water vapor enrichment preceding cloud-top brightness temperature decrease [43], [44]. Therefore, geostationary satellite temporal super-resolution modeling requires extra processing of multichannel to make use of their complementary evolution information, in contrast to VFI approaches that apply uniform optical flow across RGB channels. Second, geostationary satellite cloud data presents complex nonlinear evolution characteristics. The changes in video frames as shown in Fig. 1(c), typically comprise static backgrounds (buildings) and moving foregrounds (vehicles) with approximately linear, spatiotemporally continuous motion due to short temporal intervals, and the occlusion effects generated by foreground object motion are typically processed through occlusion coefficients as accompanying processes to uniform motion, this allows optical flow estimation using adjacent-frames under small motion assumption. However, as shown in Fig. 1(d), geostationary satellite cloud data exhibits extended temporal intervals with complex atmospheric processes affecting cloud and water vapor dynamics. These dynamics, which are influenced by atmospheric circulation, cannot be approximated as linear small motion and violate basic optical flow assumptions. In addition, the foreground changes of geostationary satellite

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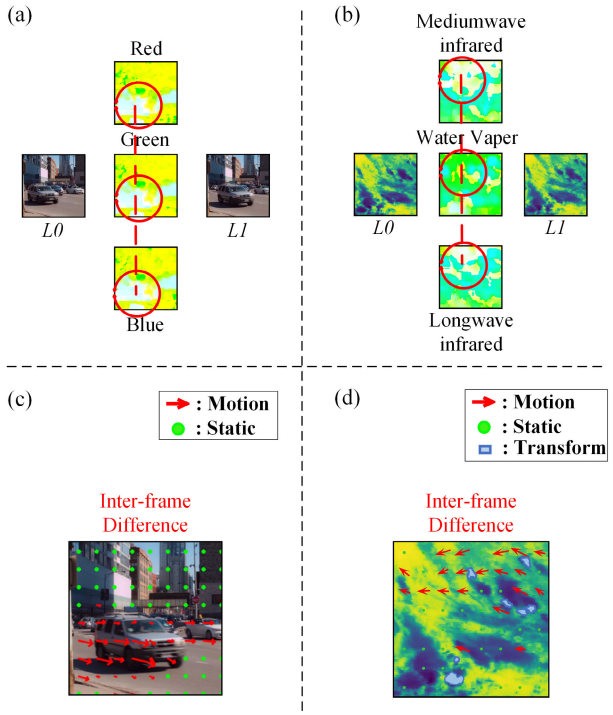


Fig. 1. Comparison of characteristics between geostationary satellite cloud data and video data. (a) Synchronized variation characteristics of the RGB channels in video data. (b) Multichannel asynchronous variation characteristics of geostationary satellite cloud data. (c) Approximate linear motion variation characteristics of video data. (d) Complex nonlinear variation characteristics of geostationary satellite cloud data.

cloud data encompass complex nonlinear processes, including the generation and dissipation of cloud and water vapor [45], [46], requiring information from multiple frames beyond adjacent frames, and background occlusion effects from nonlinear foreground change cannot be represented by simple occlusion coefficients. These backgrounds, typically composed of the Earth's surface, ocean surfaces, and other elements, which are in a relatively static state, should be handled differently.

Third, training and evaluation strategy. The goals of geostationary satellite temporal super-resolution differ fundamentally from VFI goals, requiring adapted training objectives and evaluation metrics. While video interpolation aims to generate multiple intermediate frames for visual smoothness, satellite temporal super-resolution focuses on reconstructing unobserved convective evolution processes, requiring intermediate frames to serve as reliable anchors for critical physical transitions. The sparse spatial distribution of convective regions poses challenges during model training, as these crucial features risk being smoothed out. Therefore, extra attention must be given to convective regions during the training of geostationary satellite temporal super-resolution models. Furthermore, image reconstruction metrics used in VFI frameworks inadequately quantify model performance in reconstructing convective processes, highlighting the need for specialized evaluation metrics focused on convective processes reconstruction capabilities.

To address these challenges, we propose a novel spatiotemporal self-supervised learning framework for geostationary satellite temporal super-resolution. This framework enhances the VFI techniques previously adopted in geostationary satellite cloud data temporal super-resolution tasks through comprehensive improvements in data construction, model architecture, and training-evaluation strategy. Specifically, we propose multichannel encoding and temporal-flow evolution network for optimized-observation reconstruction (METEOR-Net), a network that maintains the advantages of flow-based methods while eliminating dependence on optical flow to mitigate the incompatibility issues of flow-based approaches with geostationary satellite cloud data. The main contributions are as follows.

1) The spatiotemporal consistency between regional observation and full-disk observation of geostationary satellites is used to construct reliable temporal low-resolution and high-resolution data pairs. This method enables the formation of ground truth labels for spatiotemporal self-supervised learning constrained by spatiotemporal consistency relationships, eliminating the requirement for additional annotations.
2) METEOR-Net is proposed as a novel architecture that goes beyond previous flow-based methods through temporal sequence-flow modeling. The network introduces three synergistic modules specifically designed to address the challenges of multichannel asynchrony and complementarity, nonlinear dynamics, and the accompanying occlusion modeling.
3) A convective region-weighted loss function is proposed for training the model to make additional focus on the convective regions. Furthermore, meteorological evaluation metrics targeting convective process reconstruction are introduced to evaluate the model's ability to reconstruct convective processes.
4) Experimental results on the FY-4A satellite cloud data temporal super-resolution reconstruction task demonstrate that METEOR-Net outperforms existing temporal super-resolution methods for geostationary satellite data in both visual fidelity and convective process reconstruction accuracy, establishing it as the current state-of-the-art (SOTA) method.

II. Data

A. Introduction to FY-4A Satellite Cloud Data

The geostationary satellite cloud data used in this study is acquired from FY-4A, China's next-generation geostationary satellite launched in December 2016 [7]. The Advanced Geostationary Radiation Imager (AGRI) onboard FY-4A captures data across 14 spectral channels, spanning visible, near-infrared, mediumwave infrared, water vapor, and longwave infrared. This comprehensive spectral coverage enables monitoring of Earth's atmosphere, cloud systems, oceans, and land surface characteristics.

The study region encompasses central and eastern China (20.0° N–40.0° N, 104.0° E–125.0° E), where convective weather events occur frequently. This region is selected to ensure the temporal super-resolution results encompass numerous rapid convective processes for model validation. To maintain focus on convective events, channels 7–14 are

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TABLE I
FY-4A GEOSTATIONARY SATELLITE CHANNELS

Channel	Central Wavelength (μm)	Spatial Resolution (km)	Channel Type
7	3.75 (high)	2 km	Mediumwave infrared
8	3.75 (low)	4 km	Mediumwave infrared
9	6.25	4 km	Water vapor
10	7.1	4 km	Water vapor
11	8.5	4 km	Longwave infrared
12	10.7	4 km	Longwave infrared
13	12.0	4 km	Longwave infrared
14	13.5	4 km	Longwave infrared

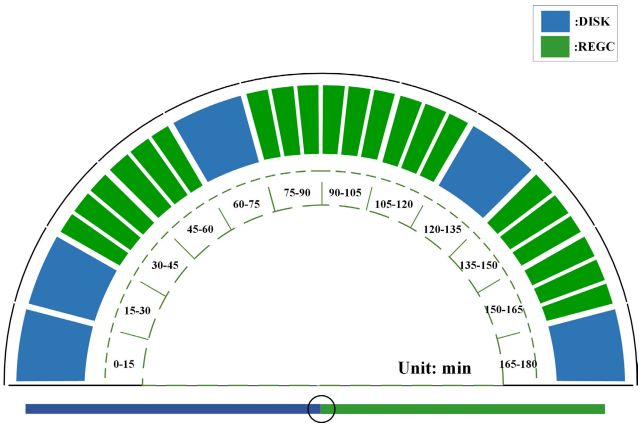


Fig. 2. FY-4A observation modes for each cycle. Blue represents full-disk observation every 15 min, and green represents regional observation three times every 15 min.

selected for experimental analysis due to their strong correlation with convective weather and continuous availability throughout the day. The corresponding channel parameters are presented in Table I.

B. Temporal Low-Resolution and High-Resolution Data Pairs

The FY-4A satellite performs full-disk observations every 15 min to provide full-disk data. This temporal resolution proves insufficient for monitoring rapid convective processes and other fast-evolving weather processes. To address this limitation, FY-4A implements a regional coverage (REGC) observation mode, which conducts three consecutive observations within a 15-min interval to provide regional data equivalent to 5-min temporal resolution. This observation mode enables high-frequency monitoring of target regions, providing crucial data support for tracking and warning convective events. However, FY-4A does not operate continuously in REGC mode. As illustrated in Fig. 2, the satellite alternates between full-disk and regional observation modes within a 3-h operational cycle.

To meet the requirement for continuous high temporal resolution coverage, an optimal approach involves fitting low-to-high resolution mapping relationships from full-disk data, thereby enhancing the temporal resolution of full-disk data. This constitutes the primary objective of the geostationary satellite temporal super-resolution tasks. The establishment of such mapping relationships requires explicit construction

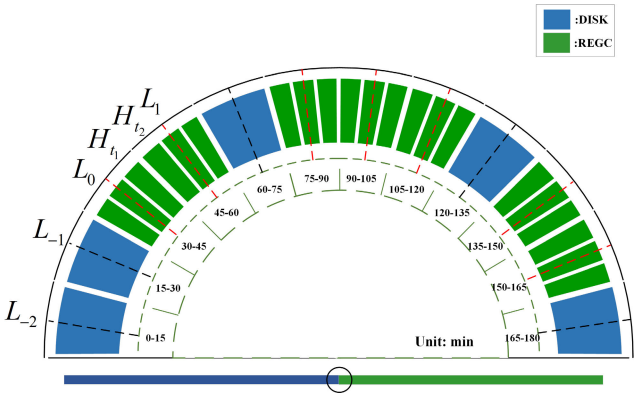


Fig. 3. Sampling mode after spatiotemporal consistency validation for FY-4A geostationary satellite observations. The black dashed line represents the original full-disk observation, and the red dashed line represents the regional observation that replaces the full-disk observation after spatiotemporal consistency validation.

of low-to-high resolution data pair representations. Given the absence of high temporal resolution records in full-disk observations, the direct construction of temporal low-resolution and high-resolution data pairs from full-disk data becomes infeasible. To overcome this constraint, a mapping relationship construction method is proposed. This method leverages the spatiotemporal consistency between full-disk data and regional data, using temporal low-resolution and high-resolution data pairs from regional data to represent full-disk temporal resolution mapping relationships. This enables the construction of low-to-high resolution mapping relationship samples in the form of data pairs.

Specifically, the foundation for using regional data in place of full-disk data to construct data pairs lies in the spatiotemporal consistency between the selected full-disk data and regional data. Spatially, both regional and full-disk observation modes use the same AGRI instrument, ensuring continuity and consistency of observational data within identical spatial domains. This instrumental consistency validates the spatial correlation between full-disk data and regional data within specified latitude–longitude bounds. Temporally, AGRI performs north-to-south scanning of the globe within each full-disk revisit period. For the study region located in the mid-to-low latitudes of the Northern Hemisphere, the scanning time of this area in full-disk coverage observations mode approximates the middle of the scanning period. Correspondingly, the intermediate temporal point of regional data during this period demonstrates strong temporal alignment with full-disk data. Following this spatiotemporal consistency validation, regional data at intermediate time points within the study area are considered spatiotemporally consistent with full-disk data for each scanning period.

As shown in Fig. 3, after spatiotemporal consistency verification, a temporal low-resolution and high-resolution data pair consists of four frames L_0, H_{t_1}, H_{t_2}, and L_1. The regional data marked by the adjacent red dashed lines serve as the temporal low-resolution input frames L_0 and L_1, where the subscripts 0–1 denote a full-disk revisit interval of 15 min. Two sets of regional data between these adjacent red dashed lines constitute the temporal high-resolution ground truth labels H_{t_1}

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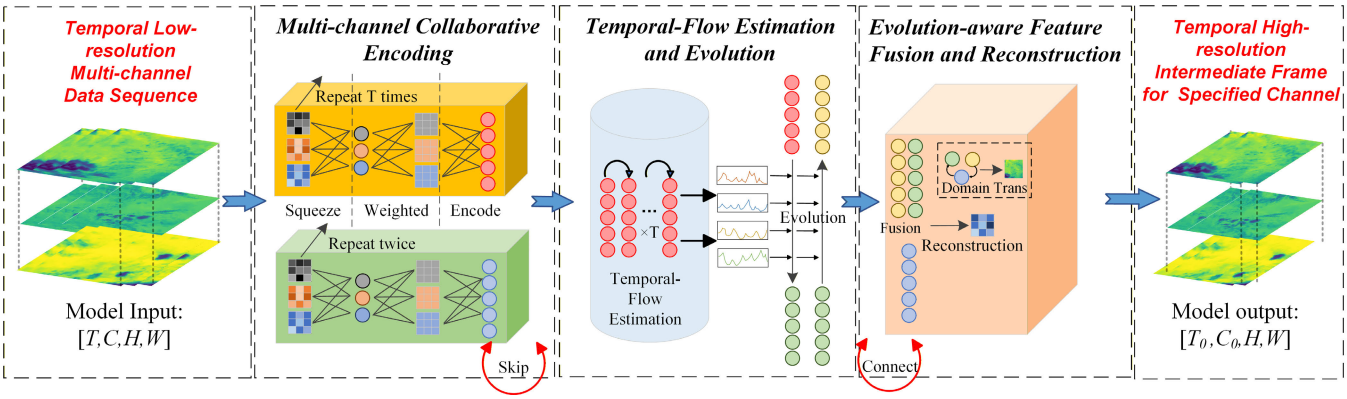


Fig. 4. Overview of the proposed METEOR-Net. It consists of the MCE module, TFEE module, and EFFR module, following an encode-evolve-reconstruct architecture.

and H_{t_2} correspond to two equally spaced time points within the 15-min full-disk revisit interval, where the subscripts 0 to t_1 to t_2 to 1 is equivalent to adding 5, 10, and 15 min to the initial time point (5-min temporal resolution).

Considering the complex evolution dynamics in geostationary satellite cloud data, the proposed temporal super-resolution model adopts a sequence-flow modeling paradigm instead of the conventional adjacent-frame optical flow-based modeling paradigm to capture higher order temporal dynamics. The input sequence is accordingly expanded by incorporating two additional frames preceding the adjacent frames, forming a six-frame sequence $L_{-2}, L_{-1}, L_0, H_{t_1}, H_{t_2}$, and L_1. Here, L_{-2}, L_{-1}, L_0, and L_1 serve as temporal low-resolution input sequence, while H_{t_1} and H_{t_2} serve as temporal high-resolution ground truth labels.

Models trained on data constructed through this method effectively represent the mapping relationship from 15- to 5-min temporal resolution in regional observations. Through spatiotemporal consistency transfer, this mapping relationship is applicable to full-disk observations, enabling the generation of temporal high-resolution geostationary satellite cloud data at 5-min intervals.

III. Method

A. Preliminaries

The temporal super-resolution task aims to reconstruct temporal high-resolution intermediate frames $\{H_{t_i}\}_{i=1}^{T}$, where $t_i \in (0, 1)$, from given temporal low-resolution adjacent frames L_0 and L_1. Existing VFI modeling paradigms based on adjacent-frame optical flow estimate the intermediate motion field from frames L_0 and L_1. Under the small uniform motion assumption, optical flow at different time points t_i is estimated to perform bidirectional flow warping on L_0 and L_1, thereby reconstructing $\{H_{t_i}\}_{i=1}^{T}$. This process can be simplified as

$$F = \text{Flow}(L_0, L_1) \tag{1}$$

$$\{H_{t_i}\}_{i=1}^{T} = W(L_0, L_1, F, t_i), \quad t_i \in (0, 1) \tag{2}$$

where F represents the optical flow estimated from adjacent frames, Flow denotes the optical flow estimation model, W indicates the warping operation, and $\{H_{t_i}\}_{i=1}^{T}$ represents the set of temporal high-resolution intermediate frames generated according to resolution requirements. Due to the complex nonlinear evolution characteristics of geostationary satellite cloud data, this modeling method frequently fails to reconstruct accurate temporal super-resolution results.

To specifically address temporal super-resolution for geostationary satellite cloud data, a sequence-flow-based modeling paradigm is proposed as

$$\{H_{t_i}\}_{i=1}^{T} = M\left(\{L_t\}_{t=-a}^{b}\right), \quad t_i \in (0, 1) \tag{3}$$

where $\{L_t\}_{t=-a}^{b}$ represents a sequence of temporal low-resolution frames, with indices a and b determined by the required sequence length. M denotes a neural network parameterized by θ. θ is optimized through gradient descent to maximize the likelihood of generating temporal high-resolution intermediate ground truth frames $\{H_{t_i}\}_{i=1}^{T}$ given the low-resolution frame sequence $\{L_t\}_{t=-a}^{b}$ as follows:

$$\theta^* = \arg\max_{\theta} P\left(\{H_{t_i}\}_{i=1}^{T} \mid \{L_t\}_{t=-a}^{b}; \theta\right). \tag{4}$$

B. Multichannel Encoding and Temporal-Flow Evolution Network for Optimized-Observation Reconstruction

The METEOR-Net implements the sequence-flow-based modeling paradigm through a parameterized neural network M. As illustrated in Fig. 4, METEOR-Net adopts an encode-evolve-reconstruct architecture to maintain sequence-flow participation in intermediate frame reconstruction without relying on optical flow estimation. The architecture is composed of three components: multichannel collaborative encoding (MCE) module, temporal-flow estimation and evolution (TFEE) module, and evolution-aware feature fusion and reconstruction (EFFR) module. The MCE module performs multichannel collaborative feature fusion and encodes the input sequence into latent space. Subsequently, the TFEE module estimates multiscale temporal-flows and guides high-resolution spatiotemporal evolution. The EFFR module integrates evolved spatiotemporal features with pre-evolution spatial distribution features in the original space to reconstruct intermediate frames through feature redistribution.

C. MCE Module

Unlike RGB channels in video data, the channels in geostationary satellite cloud data have distinct physical significance

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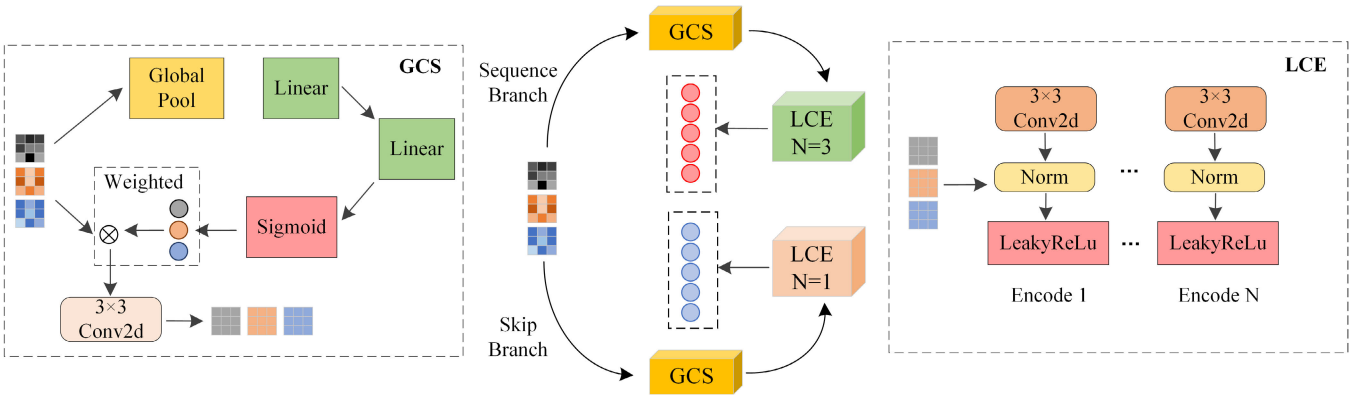


Fig. 5. Structure of the MCE module. It is divided into a full sequence branch and a skip branch, each consisting of GCS and local convolution encoding (LCE).

and exhibit asynchronous spatiotemporal evolution characteristics. These characteristics enable complementary information exchange across channels, where temporal evolution in one channel can compensate for information gaps in others. For instance, water vapor variations typically precede and serve as precursors to changes in cloud-top brightness temperature. However, previous geostationary satellite cloud data temporal super-resolution methods have not adequately addressed these characteristics, resulting in a lack of spatiotemporal consistency constraints between channels and the utilization of collaborative information during the feature encoding process. To address these limitations, the MCE module, inspired by channel attention mechanisms [47], is designed to perform compressed encoding of multichannel geostationary satellite cloud data.

As illustrated in Fig. 5, the MCE module consists of two parallel branches, each consisting of a global channel squeeze (GCS) submodule and a local convolutional encoding submodule. The GCS submodule operates in the original data space, performing channel squeeze on each multichannel frame to generate channel descriptors based on global information. These descriptors learn sample-specific activation through a channel-dependent self-gating mechanism, selectively emphasizing and suppressing channel features. This differentiable process enables the model to adjust channel weights in the fused features, following the computation as:

$$s = \sigma\left(W_{s2} * \delta\left(W_{s1} * \sum_{i=1}^{H}\sum_{j=1}^{W}\frac{L_t(i,j)}{HW} + b_{s1}\right) + b_{s2}\right) \quad (5)$$

$$\tilde{L}_t = W_{xs} * (L_t \odot s) \quad (6)$$

where $L_t \in \mathbb{R}^{C_{\text{in}}\times H\times W}$ denotes the module input, $s \in \mathbb{R}^{C_{\text{in}}}$ represents the channel descriptors generated from global information, and $\tilde{L}_t \in \mathbb{R}^{C_{\text{out}}\times H\times W}$ indicates the multichannel fused features after gating-based weighting. W_{s1} and W_{s2} denote 1×1 convolutional matrices for affine transformation, W_{xs} represents a 3×3 convolutional kernel for local weighted summation, b_{s1} and b_{s2} are bias terms, $\delta(\cdot)$ denotes the LeakyReLU activation function, and $\sigma(\cdot)$ represents the Sigmoid activation function.

The local convolutional encoding submodule follows the GCS submodule in series, performing nonlinear spatial transformations on the compressed and fused data through multiple small local convolutional kernels combined with normalization and activation functions. This process accomplishes dimensionality reduction and mapping from the original data space to the latent space. The resulting latent space features exhibit enhanced abstraction and information density while achieving feature dimension compression and redundancy reduction to minimize computational overhead. The process is formulated as

$$f\left(\tilde{L}_t\right) = \delta\left(\text{GN}\ \left(W * \tilde{L}_t + b\right)\right) \quad (7)$$

$$X_t = f_N \circ f_{N-1} \circ \cdots \circ f_1\left(\tilde{L}_t\right) \quad (8)$$

where $f(\tilde{L}_t)$ represents single-layer local convolutional encoding computation, $X_t \in \mathbb{R}^{C_{\text{out}}\times H''\times W''}$ denotes the compressed encoded feature sequence, W represents the convolutional matrix, b indicates the bias term, GN denotes group normalization, and $\circ$ denotes function composition. The MCE module operates two parallel branches at the model's preinput stage. The sequence branch takes the temporal low-resolution sequence $\{L_t\}_{t=-a}^{b}$ as input, compressing and encoding it into feature sequence $\{X_t\}_{t=-a}^{b}$. The skip branch, taking adjacent frames L_0, L_1 as input, generates skip features $\{X'_{t_i}\}_{i=1}^{T}$. The skip branch is designed with a single-layer local convolution to maximally preserve invariant features from the original space and ensure the subsequent reuse of spatial information from skip features.

D. TFEE Module

The previous optical flow-based VFI paradigm applied to video data typically relies on small motion assumptions, treating motion as uniform optical flow between adjacent frames due to short temporal intervals and small frame-to-frame variations. This method enables motion estimation and image warping based on the optical flow paradigm. However, geostationary satellite cloud data presents fundamentally different characteristics, including extended temporal intervals and significant frame-to-frame variations. Water vapor and cloud dynamics, influenced by complex atmospheric circulation, frequently exhibit nonlinear evolution processes, including convective generation and dissipation between frames. These inherent characteristics render small uniform motion assumptions inadequate for geostationary

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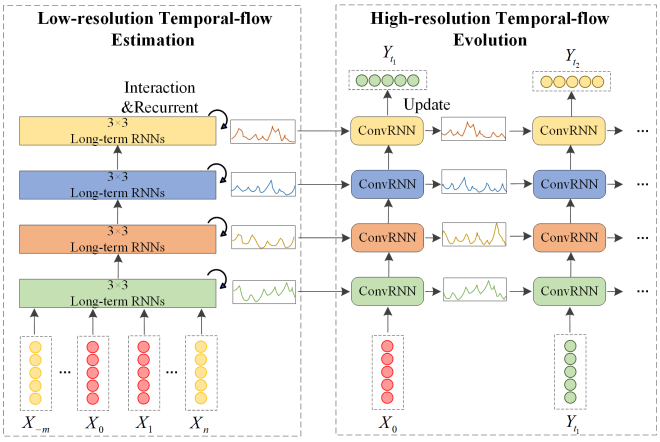


Fig. 6. Structure of the TFEE module. It consists of a low-resolution temporal-flow estimation module and a high-resolution temporal-flow evolution module.

satellite cloud data modeling. Under nonuniform motion conditions, the description of continuous motion through finite points presents inherent limitations: two points can only define linear change, failing to capture quadratic variations. Consequently, adjacent-frame information alone proves insufficient for characterizing higher order nonlinear evolution between frames [48].

The TFEE module addresses these complex, nonuniform dynamics through comprehensive design considerations. By using compressed encoded temporal low-resolution sequences rather than adjacent frames as input, the module models temporal characteristics within the sequence and generates multiscale temporal-flows as implicit variations representations to incorporate higher order nonlinear evolution information from extended sequences. The module implements a spatiotemporal evolution method to simulate cloud system generation, dissipation, and movement processes under temporal-flow guidance, replacing conventional optical flow warping. This design enables better fitting of nonlinear evolution processes while avoiding the limitations imposed by optical flow assumption inductive bias on model capabilities.

As illustrated in Fig. 6, the TFEE module architecture consists of two submodules: the low-resolution temporal-flow estimation module and the high-resolution temporal-flow evolution module. The module processes low-resolution feature sequences $\{X_t\}_{t=-a}^{b}$ encoded by the MCE module as input. During each iteration, the initialized hidden state Z_{t-1}^n interacts with current temporal low-resolution sequence features through a long-range convolutional recurrent module. The low-resolution sequence spatiotemporal interaction process follows as:

$$\left[i_t^n, f_t^n, o_t^n, \tilde{c}_t^n\right] = \phi\left(\mathrm{LN}\left(W_x^n * X_t^n + W_h^n * Z_{t-1}^n\right) + b^n\right) \tag{9}$$

$$C_t^n = f_t^n \odot C_{t-1}^n + i_t^n \odot \tilde{c}_t^n \tag{10}$$

$$Z_t^n = o_t^n \odot \phi\left(C_t^n\right) \tag{11}$$

where $X_t^n \in \mathbb{R}^{C_{\text{out}} \times H'' \times W''}$ denotes low-resolution features, $Z_{t-1}^n \in \mathbb{R}^{C_{\text{hid}} \times H'' \times W''}$ represents the initialized hidden state, $C_t^n \in \mathbb{R}^{C_{\text{hid}} \times H'' \times W''}$ indicates the memory state, W_x^n and W_h^n represent convolution matrices associated with input and hidden state calculations, b^n denotes bias terms, LN represents configurable layer normalization, $\phi(\cdot)$ denotes an abstract activation function, and i_t^n, f_t^n, o_t^n, and $\tilde{c}_t^n$ indicate four gating terms in the recurrent process. The superscript n denotes the nth layer module, while the subscript t indicates the tth moment in recurrent computation.

Through spatiotemporal interaction calculations across different layers in the low-resolution temporal-flow estimation module, the network receptive field expands with increasing layer depth, accompanied by enhanced nonlocality of spatiotemporal features. This enables the establishment of temporal-flows at different scales through recurrent operations across layers. The process is formulated as

$$\left\{F_{low}^n\right\}_{n=1}^{N} = \mathcal{F}_{low}\left(\{X_t\}_{t=-a}^{b}; \left\{W_{xl}^n, W_{hl}^n, b_l^n\right\}_{n=1}^{N}\right) \tag{12}$$

where $\mathcal{F}_{low}$ represents the low-resolution sequence spatiotemporal interaction process, $\{X_t\}_{t=-a}^{b} \in \mathbb{R}^{C_{\text{out}} \times H'' \times W''}$ denotes the low-resolution feature sequence, $\{W_{xl}^n, W_{hl}^n, b^n\}_{n=1}^{N}$ encompasses all convolution matrices and bias terms associated with low-resolution sequence temporal-flows computation, and $\{F_{low}^n\}_{n=1}^{N} \in \mathbb{R}^{C_{\text{hid}} \times H'' \times W''}$ represents the generated multiscale temporal-flows. Notably, given the distinct locality characteristics of convective weather as the research focus, recurrent convolutions employ small convolution kernels to prevent excessive smoothing of small-scale local features.

The high-resolution temporal-flow evolution module uses multiscale temporal-flows as initial constraints to generate evolved features, constituting the core super-resolution mechanism of the model. While the previous optical flow-based interpolation paradigm generates intermediate frames by estimating optical flow from adjacent frames and then warping the frames accordingly, this module replaces the warping process with temporal-flow evolution to generate intermediate features as

$$\left\{F_{high}^n\right\}_{n=1}^{N}, \left\{Y_{t_i}\right\}_{i=1}^{T} = \mathcal{F}_{high}\left(X_0, \left\{F_{low}^n\right\}_{n=1}^{N}; \left\{W_{xh}^n, W_{hh}^n, b_h^n\right\}_{n=1}^{N}\right) \tag{13}$$

where $\mathcal{F}_{high}$ represents the high-resolution temporal-flow evolution process, $X_0 \in \mathbb{R}^{C_{\text{out}} \times H'' \times W''}$ denotes the initial input at t = 0, W_{xh}^n, W_{hh}^n denote all convolution matrices associated with high-resolution temporal-flow evolution computation, $\{F_{high}^n\}_{n=1}^{N} \in \mathbb{R}^{C_{\text{hid}} \times H'' \times W''}$ indicates the continuously updated high-resolution temporal-flow, and $\{Y_{t_i}\}_{i=1}^{T} \in \mathbb{R}^{C_{\text{out}} \times H'' \times W''}$ represents the evolved feature sequence in latent space.

The module initializes evolution with sequence features X_0 at t = 0, providing initial spatial distribution characteristics. The high-resolution temporal-flow evolution process $\mathcal{F}_{high}$ symmetrically corresponds to the low-resolution sequence spatiotemporal interaction process $\mathcal{F}_{low}$. During this process, multiscale temporal-flows $\{F_{low}^n\}_{n=1}^{N}$ are introduced in stages, generating spatiotemporal features indicating cloud movement, generation, and dissipation at different scales through recurrent convolution calculations. The intermediate evolved features Y_{t_1} at t_1 are produced through dimensional reduction via 1×1 convolution fusion. Y_{t_1} serves as the initial input for subsequent evolution, repeating the spatiotemporal interaction evolution process to generate all intermediate evolved features according to temporal resolution requirements, thus completing the generation of evolved feature sequence $\{Y_{t_i}\}_{i=1}^{T} \in \mathbb{R}^{C_{\text{out}} \times H'' \times W''}$.

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During temporal-flow evolution, the representation of the variation field $\{F_{high}^{n}\}_{n=1}^{N}$ no longer changes linearly with time but updates at fixed temporal anchors. Each super-resolution temporal anchor updates the variation field to simulate nonuniform evolution processes. The temporal-flow evolution process approximates spatiotemporal prediction, with the distinction that the constraint information is established through multiscale temporal-flow evolution with known initial and final states $\{X_t\}_{t=-a}^{b}$. While spatiotemporal prediction forecasts future states based on historical information, temporal super-resolution reconstruction reconstructs the intermediate processes based on known future states, essentially implementing an inverse application of spatiotemporal prediction models through the TFEE module.

E. EFFR Module

The temporal low-resolution sequence $\{L_t\}_{t=-a}^{b}$ is mapped to latent space through the MCE module and undergoes temporal-flow evolution in the TFEE module to generate evolved features $\{Y_{t_i}\}_{i=1}^{T}$. As these evolved features exist in latent space, the EFFR module uses a decoding structure approximately symmetric to the MCE module to align the dimensions of latent space evolved features $\{Y_{t_i}\}_{i=1}^{T}$ with pre-evolution skip features $\{X'_{t_i}\}_{i=1}^{T}$. These features are then fused to reconstruct temporal high-resolution intermediate frames. The skip features, representing pre-evolution spatial distribution characteristics extracted through shallow networks from the original data, are used to account for relatively static background regions in 3-D space, such as land surfaces, ocean surfaces, and low-level clouds or fog unaffected by weather systems. These relatively low-level regions are susceptible to varying degrees of occlusion by evolving upper-layer cloud systems. The pre-evolution spatial distribution characteristics represented by skip features provide a rapid feature memory highway for reconstructing these relatively static regions after occlusion, thereby balancing potentially inappropriate evolution computations in specific regions.

The EFFR module architecture, illustrated in Fig. 7, consists of a reconstruction module and an evolution-aware fusion module. The reconstruction module incorporates alternating convolutional and deconvolutional layers to restore dimensionality compressed by the MCE module as

$$\tilde{Y}_{t_i} = \delta\left(GN\left(W_2^T * \delta\left(\mathrm{GN}\ \left(W_1 * Y_{t_i} + b_1\right)\right) + b_2\right)\right) \tag{14}$$

where $\tilde{Y}_{t_i} \in \mathbb{R}^{C_{\text{out}} \times H \times W}$ denotes dimension-aligned evolved features, $Y_{t_i} \in \mathbb{R}^{C_{\text{out}} \times H'' \times W''}$ represents evolved features in latent space, W_1 and W_2^T indicate convolution and transposed convolution matrices, respectively, b denotes the bias terms, and GN denotes group normalization. The evolved feature sequence $\{Y_{t_i}\}_{i=1}^{T}$ is sequentially input to the reconstruction module to obtain the dimension-aligned evolved feature sequence $\{\tilde{Y}_{t_i}\}_{i=1}^{T}$.

The evolution-aware fusion module processes dimension-aligned evolved features $\tilde{Y}_{t_i}$ and skip features X'_{t_i}. These inputs are added along the channel dimension to form summed features $I_{t_i} \in \mathbb{R}^{C_{\text{out}} \times H \times W}$ as

$$I_{t_i} = X'_{t_i} + \tilde{Y}_{t_i}. \tag{15}$$

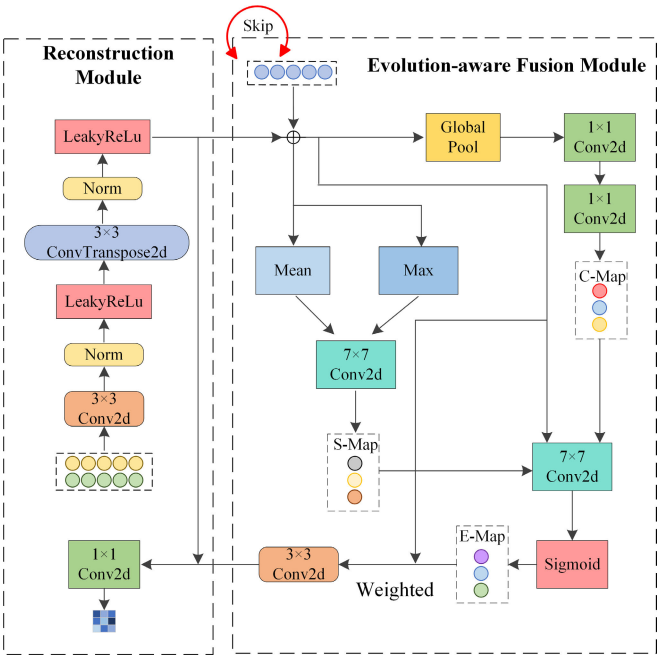


Fig. 7. Structure of the EFFR module. It consists of the reconstruction module and the evolution-aware fusion module.

The summed features undergo evolution-aware map computation through channel and spatial decoupling attention blocks as

$$c_{map} = W_{c_2} * \delta\left(W_{c_1} * \frac{\sum_{h=1}^{H}\sum_{w=1}^{W} I_{t_i}(h,w)}{HW} + b_{c_1}\right) + b_{c_2} \tag{16}$$

$$s_{map} = W_s * \mathrm{concat}\left(\frac{1}{C_{\text{out}}}\sum_{c=1}^{C_{\text{out}}} I_{t_i}(c), \max_c\left(I_{t_i}\right)\right) + b_s \tag{17}$$

$$e_{map} = \sigma\left(W_p * \mathrm{concat}\left(c_{map} + s_{map}, I_{t_i}\right) + b_p\right) \tag{18}$$

where c_{map}, s_{map}, and e_{map} represent channel weight map, spatial weight map, and evolution-aware map, respectively, W denotes convolution matrices, b indicates bias terms, and max performs maximum value selection along channels. The evolution-aware map mediates the tradeoff between pre-evolution skip features representing original spatial distribution characteristics and evolved features that have undergone both deep encode–decode processing and temporal-flow evolution.

The evolution-aware map performs balanced fusion of pre-evolution and evolved features, restoring original spatial feature information when the model requires it, thereby ensuring spatial completeness and granularity in the reconstructed intermediate frames. The corresponding process is as follows:

$$\tilde{I}_{t_i} = I_{t_i} + e_{map} \cdot \tilde{Y}_{t_i} + \left(1 - e_{map}\right) \cdot X'_{t_i}. \tag{19}$$

The fused features $\tilde{I}_{t_i} \in \mathbb{R}^{C_{\text{out}} \times H \times W}$ undergo final scale restoration to reconstruct intermediate frames $\hat{H}_{t_i} \in \mathbb{R}^{C_{\text{final}} \times H \times W}$ in data space as follows:

$$\hat{H}_{t_i} = W_{h1} * \left(W_{h2} * \tilde{I}_{t_i} + b_h\right) \tag{20}$$

where W and b represent scale restoration convolution matrices and bias terms and W_{h2} includes GN and activation.

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The design of the EFFR module is inspired by content-guided attention in dehazing models [49], where weight map fusion of high and low-level features enhances feature interaction during image reconstruction. This characteristic is adapted for super-resolution modeling, enabling evolution-aware fusion of pre-evolution and evolved features during intermediate frame pixel reconstruction, thereby strengthening the integration of original spatial and temporal evolution features to enhance reconstruction accuracy and granularity.

F. Convective Region-Weighted Loss Function

Loss functions quantify the difference between model predictions and ground truth values, serving as objective functions for optimization during model training. In temporal super-resolution tasks, loss functions typically measure the difference between reconstructed intermediate frame sequences and high-resolution ground truth label sequences. As a reconstruction-based regression task, the temporal super-resolution task conventionally uses regression losses such as mean absolute error (MAE). For video frames, standard regression losses effectively address global prediction errors, ensuring interframe continuity and smoothness through the minimization of overall prediction error, aligning with VFI objectives. However, for geostationary satellite cloud data, the primary objective of temporal super-resolution lies in reconstructing convective processes unobserved by temporal low-resolution observations. Given that convective regions, which constitute the primary areas of interest, typically occupy only a minimal portion of the data, direct application of MAE loss functions biases model learning toward global smoothness while neglecting intense variations in localized convective regions. Consequently, the loss function requires modification to make additional focus on convective regions during model training.

For model robustness considerations, the MAE loss is selected as the foundational loss function

$$l\left(H,\hat{H}\right)=\frac{1}{n}\sum_{i=1}^{n}\left|H_i-\hat{H}_i\right| \tag{21}$$

where H_i denotes the ith pixel of the high-resolution intermediate ground truth frame, $\hat{H}_i$ represents the ith pixel of the temporal high-resolution intermediate frame, and n indicates the total number of pixels.

Building upon the MAE loss, weighted attention for convective regions is introduced

$$l_r\left(H,\hat{H}\right)=\frac{1}{n}\sum_{i=1}^{n}w\left(H_i\right)\left|H_i-\hat{H}_i\right| \tag{22}$$

$$w\left(H_i\right)=\begin{cases}\alpha, & H_{i,10.7\mu\mathrm{m}}\geq T_1\\ \beta, & T_2\leq H_{i,10.7\mu\mathrm{m}}<T_1\\ \gamma, & H_{i,10.7\mu\mathrm{m}}<T_2\end{cases} \tag{23}$$

where T_1 and T_2 represent the thresholds for convective and deep convective regions, respectively, α, β, and γ represent weighted coefficients and $H_{i,10.7\mu\mathrm{m}}$ represents the 10.7 μm channel of the high-resolution intermediate ground truth frame. Based on research in convective region identification [50], areas with brightness temperature below 241K in the 10.7 μm channel are identified as potential convective cloud regions, while areas below 221K indicate strong convective activity. Accordingly, T_1 is set to 241K and T_2 to 221K.

The weighted single-step loss is accumulated across temporal super-resolution steps to form the final loss function

$$l_{wr}=\frac{1}{T}\sum_{i=1}^{T}l_r\left(H_{t_i},\hat{H}_{t_i}\right). \tag{24}$$

IV. Experiment

A. Dataset

The raw data for the dataset is acquired from the AGRI aboard the FY-4A satellite [7]. The dataset encompasses L1 products with 4-km spatial resolution collected from May 2018 to September 2023, covering a geographical domain of 20.00°N –40.00°N latitude and 104.00°E–125.00°E longitude.

After constructing paired samples and preprocessing sequences, we remove invalid data through continuity checks and missing data detection. NaN values in FY-4A observations are physically meaningless and usually indicate pixel dropout caused by scan or calibration anomalies. Even a few NaN pixels can produce nonphysical artifacts during normalization and convolutional propagation, which degrades training stability and reduces evaluation reproducibility. We therefore apply a strict conservative rule: a frame is marked invalid if any pixel contains NaN. For time series, temporal super-resolution training and evaluation use data pairs built from a six-frame window with fixed relative time offsets, $L_{-2}, L_{-1}, L_0, H_{t_1}, H_{t_2}$, and L_1. A missing frame at any time creates an unevenly spaced sequence and violates the temporal constraint, which introduces uncontrolled interpolation or alignment errors. To keep the input spacing consistent and ensure fair evaluation, the maximum allowed number of consecutive missing frames within a window is set to 0, and we discard the entire sequence once any frame is missing.

After the above processing, the dataset is split in an approximate ratio of 4:1:1, with data from 2018 to 2021 used for training, data from 2022 used for validation, and data from 2023 reserved as an independent test set. This "train-on-the-past, test-on-the-future" partitioning strategy is commonly adopted in temporal deep learning tasks to prevent temporal information leakage (e.g., training, validation, and test sets sharing data from the same year, which may lead to overfitting and overestimate the generalization to future scenarios) and to ensure the reliability of performance evaluation. All comparison methods use exactly the same split to guarantee a fair comparison. Each sequence $L_{-2}, L_{-1}, L_0, H_{t_1}, H_{t_2}$, and L_1 consists of six frames, comprising four temporal low-resolution input frames at 15-min intervals and two temporal high-resolution ground truth frames at 5-min intervals.

To optimize computational resources and temporal efficiency while ensuring experimental fairness and validity, centrally cropped 128 × 128 subregion blocks are used for model training and testing. This method aligns with established practices in neural network experiments involving geostationary satellite cloud data [51], [52], [53]. Subsequently, generalization experiments were conducted over

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nontraining subregion blocks located in areas with complex terrain and different climatic regimes, in order to assess the model's generalization ability and applicability in regions that were not used for training.

B. Experimental Setup

The METEOR-Net is configured with the following architectural parameters: The sequence branch of the MCE module comprises three layers, while the skip branch consists of one layer, both with 64 hidden units. The TFEE module is structured with four layers and 256 hidden units. In the EFFR module, 3×3 convolutional/deconvolutional kernels are used for the reconstruction module, while 7×7 convolutional kernels are used for evolution-aware mapping and pixel rearrangement in the fusion module. LeakyReLU [54] is used as the primary activation function in most parts of the network, except for weighting processes where Sigmoid activation is applied for normalization, and the recurrent unit, where Sigmoid and Tanh are used. In the region-weighted loss function, α, β, and γ are set to 1, 2, and 5, respectively.

To ensure experimental fairness in comparative analysis, all models used for comparison are implemented using official open-source code. Model training is conducted uniformly on NVIDIA Tesla A100 GPUs with consistent hyperparameters: batch size of 64, maximum training epochs of 200, Adam optimizer with initial learning rate of 3×10^{-4}. The learning rate decreases when the validation set loss no longer decreases, and training stops when it drops below 8×10^{-5}.

C. Evaluation Metrics

The quantitative evaluation metrics for model performance are categorized into two groups: image quality metrics and meteorological evaluation metrics. Both groups are calculated frame-wise and then averaged over all evaluated frames.

Image quality metrics include peak signal-to-noise ratio (PSNR) [55], structural similarity index (SSIM) [56], and root mean square error (RMSE). PSNR is based on mean square error (mse) between ground-truth frames and reconstruction frames, with higher values indicating smaller reconstruction deviations. SSIM quantifies structural similarity between ground truth and reconstructed frames, where higher values denote greater similarity. RMSE quantifies the pixel-level error between the reconstructed frames and the ground truth frames, where a lower value indicates smaller errors. These metrics evaluate temporal super-resolution global visual fidelity across all channels (channel 7–14), failing to assess the critical reconstruction performance of convective processes in geostationary satellite temporal super-resolution.

Meteorological evaluation metrics include critical success index (CSI), probability of detection (POD), and false alarm rate (FAR), which are commonly used in precipitation nowcasting evaluation [57], [58], [59], [60]. CSI, POD, and FAR, traditionally utilized in precipitation forecast evaluation, are adapted for geostationary satellite cloud data assessment. Since the 10.7-μm channel is situated within the infrared atmospheric window and experiences minimal atmospheric absorption, it serves as the primary channel for the identification and tracking of convective clouds. Quantitative evaluation results from the 10.7-μm channel effectively reflect the model's performance in reconstructing convective processes. Therefore, the 10.7-μm channel is selected as the channel for assessing the performance of convective reconstruction. Brightness temperature thresholds of 273K, 241K, and 221K are established to identify cold cloud, convective cloud, and strong convective cloud regions [50], respectively. Given that lower cloud-top brightness temperatures typically indicate stronger convective activity, pixels below these thresholds are designated as positive samples. This enables the computation of hits (reconstruction = 1 and truth = 1), misses (reconstruction = 0 and truth = 1), and false alarms (reconstruction = 1 and truth = 0) for subsequent CSI, POD, and FAR calculations.

CSI, also known as the threat score, integrates hit, miss, and false alarm events as

$$\mathrm{CSI} = \frac{\mathrm{Hits}}{\mathrm{Hits} + \mathrm{Misses} + \mathrm{False\ Alarm}}. \tag{25}$$

POD measures the model's ability to capture actual events as

$$\mathrm{POD} = \frac{\mathrm{Hits}}{\mathrm{Hits} + \mathrm{Misses}}. \tag{26}$$

FAR quantifies the proportion of incorrectly identified events as

$$\mathrm{FAR} = \frac{\mathrm{False\ Alarm}}{\mathrm{Hits} + \mathrm{False\ Alarm}}. \tag{27}$$

D. Multichannel Reconstruction Comparison Experiment

In this section, the performance of METEOR-Net in the multichannel temporal super-resolution reconstruction task will be evaluated and compared with traditional temporal super-resolution methods and deep learning-based temporal super-resolution methods. Traditional methods include linear interpolation and optical flow methods [61], where linear interpolation serves as a baseline using bilinear interpolation for intermediate frame generation, while the optical flow method uses the classical Farneback algorithm for dense optical flow field estimation between adjacent frames. The deep learning-based baselines include FLAVR, SSM-T, and EMAIRA-VFI models. FLAVR, proposed by Kalluri et al. [30], represents the SOTA performance in VFI algorithms. SSM-T, recently developed by Vandal and Nemani [42] for the GOES-R satellite, represents the current SOTA in temporal super-resolution of geostationary satellite cloud data. EMAIRA-VFI, proposed by Wang et al. [62] is a transformer-based temporal super-resolution model for meteorological data, representing advanced techniques in meteorological temporal super-resolution.

Table II presents quantitative evaluation results across channels 7–14 of FY-4A for METEOR-Net and comparative models. EMAIRA-VFI provides a strong Transformer-based baseline and achieves competitive performance among the learning-based methods. Nevertheless, METEOR-Net still attains the best PSNR and the lowest RMSE on all channels, and achieves the best SSIM on 7 out of the 8 channels, attributed to its architectural design specifically tailored for geostationary satellite cloud data characteristics. These

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TABLE II

QUANTITATIVE EVALUATION RESULTS OF MULTICHANNEL TEMPORAL SUPER-RESOLUTION RECONSTRUCTION

Ch	μm	PSNR ↑						SSIM ↑						RMSE ↓					
		Linear	Flow	SSM-T	FLAVR	EMAIRA-VFI	Our	Linear	Flow	SSM-T	FLAVR	EMAIRA-VFI	Our	Linear	Flow	SSM-T	FLAVR	EMAIRA-VFI	Our
7	3.75	44.97	44.75	48.32	48.01	48.65	**49.66**	0.9674	0.9685	0.9864	0.9851	0.9868	**0.9897**	1.6724	1.7012	1.0645	1.1011	1.0368	**0.9064**
8	3.75	45.44	45.18	48.77	48.45	49.30	**50.08**	0.9704	0.9713	0.9874	0.9864	0.9885	**0.9902**	1.6008	1.6376	1.0249	1.0593	0.9675	**0.8809**
9	6.25	50.79	51.14	52.39	52.92	52.94	**53.18**	0.9934	0.9930	0.9951	**0.9961**	0.9956	0.9959	0.7622	0.7415	0.6215	0.5883	0.5873	**0.5712**
10	7.1	49.81	50.01	52.50	52.77	52.98	**53.61**	0.9897	0.9888	0.9947	0.9952	0.9952	**0.9960**	0.8976	0.9036	0.6248	0.6102	0.5956	**0.5479**
11	8.5	46.71	46.62	50.70	50.47	51.35	**52.51**	0.9774	0.9791	0.9922	0.9920	0.9933	**0.9951**	1.4117	1.4136	0.8229	0.8394	0.7573	**0.6508**
12	10.7	46.16	46.12	50.16	50.29	50.68	**52.17**	0.9748	0.9773	0.9912	0.9914	0.9922	**0.9948**	1.5172	1.5003	0.8855	0.8636	0.8297	**0.6816**
13	12.0	46.20	46.08	50.06	50.23	50.66	**51.99**	0.9761	0.9780	0.9915	0.9918	0.9926	**0.9947**	1.4777	1.4765	0.8734	0.8548	0.8127	**0.6824**
14	13.5	44.28	43.74	45.63	46.20	46.26	**46.64**	0.9768	0.9741	0.9858	0.9877	0.9880	**0.9887**	1.6203	1.7267	1.3525	1.2789	1.2644	**1.2100**

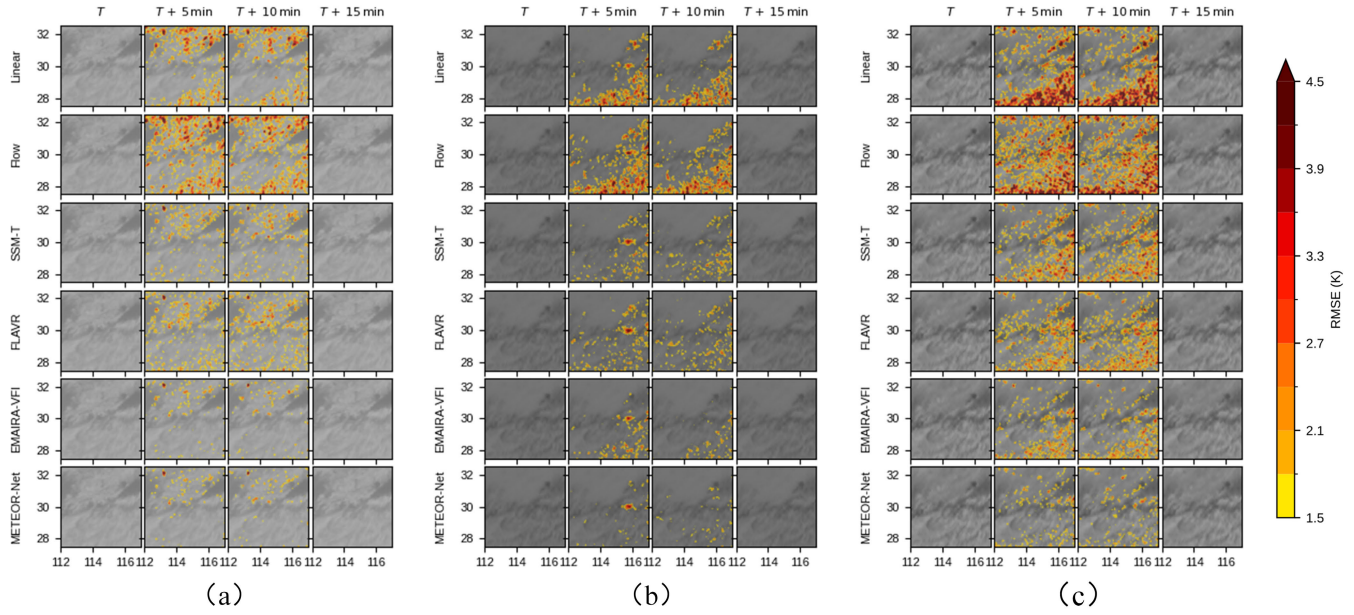


Fig. 8. Comparison of error maps for individual cases of FY-4A multichannel temporal super-resolution reconstruction. T and T + 15 min represent full-disk observations of temporal low-resolution input, while T + 5 min and T + 10 min represent the RMSE error maps between the temporal super-resolution frames and the ground truth frames. The time is from 01:45 UTC to 02:00 UTC on July 4, 2023. (a) Channel 8 of the mediumwave infrared channel. (b) Channel 10 of the water vapor channel. (c) Channel 12 of the longwave infrared channel.

quantitative results validate the model's capability for accurate temporal super-resolution reconstruction across diverse spectral channels.

Fig. 8(a)–(c) illustrates error map examples for channels 8, 10, and 12 (mediumwave infrared, water vapor, and longwave infrared, respectively). Compared to traditional linear interpolation and optical flow methods, deep learning-based methods demonstrate a significant reduction in both error extent and magnitude. However, high-error regions persist in areas of substantial variation. METEOR-Net, through its nonlinear dynamic modeling of channel temporal-flow evolution, further reduces the occurrence of high-error points.

E. Convective Processes Reconstruction Comparison Experiment

The 10.7-μm channel of the FY-4A geostationary satellite is located within an atmospheric window with minimal absorption, enabling its channel brightness temperature to effectively represent the cloud-top temperature of convective clouds. Consequently, this channel is commonly utilized for the identification and tracking of convective clouds. Thresholds of 273K, 241K, and 221K [50] are applied to the 10.7-μm channel to calculate CSI, POD, and FAR, thereby quantitatively evaluating the model's temporal super-resolution reconstruction performance in cold cloud regions, convective cloud regions, and strong convective cloud regions. These quantitative evaluation results are used to indicate the model's performance in reconstructing convective processes.

Fig. 9 illustrates the variation in CSI, POD, and FAR across these three brightness temperature thresholds for six temporal super-resolution reconstruction methods, with metrics averaged across all test samples. METEOR-Net consistently achieves superior CSI and POD values across all thresholds while maintaining low FAR, and exhibits notably slower degradation with decreasing brightness temperature thresholds compared to alternative methods. This performance characteristic indicates robust capability of METEOR-Net in fitting rapid convective evolution processes within convective and strong convective cloud regions. While competing methods achieve comparable CSI and POD values at the 273K threshold, their performance deteriorates rapidly at lower thresholds, suggesting limited capability in modeling nonlinear processes such as convective cloud generation, development, merger, and dissipation within convective regions.

Fig. 10 presents two case studies of convective evolution. Fig. 10(a) depicts a typical early stage deep convective cloud

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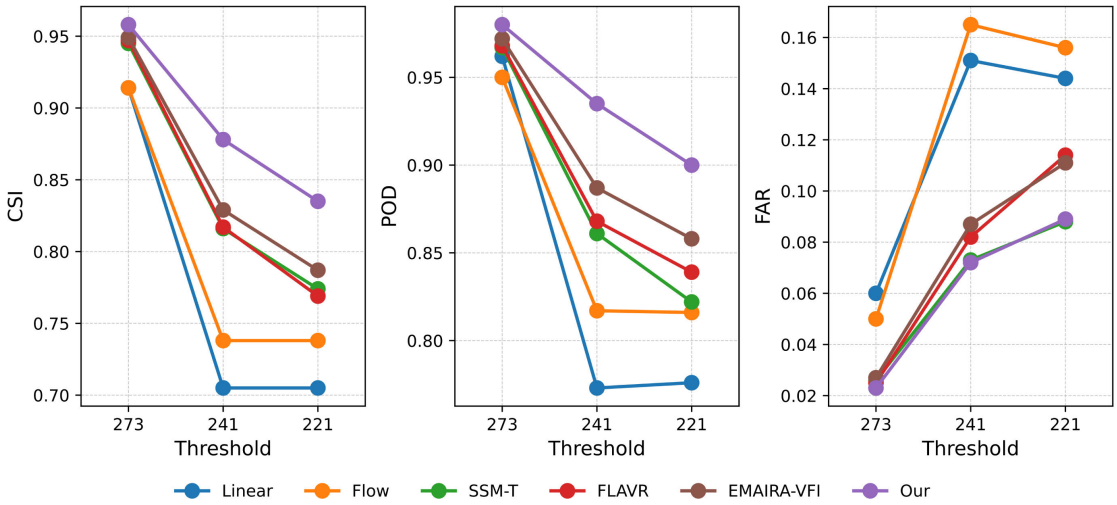


Fig. 9. Comparison of the changes in CSI, POD, and FAR metrics of temporal super-resolution reconstruction results for different models with decreasing brightness temperature thresholds.

TABLE III
QUANTITATIVE RESULTS OF MULTISCALE INPUT EXPERIMENTS WITH DIFFERENT INPUT PATCH SIZES

Input size	SSIM ↑	PSNR ↑	RMSE ↓	CSI ↑			POD ↑			FAR ↓		
				T<273	T<241	T<221	T<273	T<241	T<221	T<273	T<241	T<221
128 × 128	0.9948	52.17	0.6816	0.958	0.878	0.835	0.980	0.935	0.900	0.023	0.072	0.089
64 × 64	0.9944	52.01	0.6997	0.958	0.870	0.828	0.979	0.932	0.897	0.023	0.079	0.098
32 × 32	0.9940	51.77	0.7220	0.957	0.867	0.824	0.980	0.930	0.896	0.026	0.080	0.103

formation process. The process involves cloud-top center ascent to the tropopause accompanied by cooling. While the convective evolution process is not captured in temporally low-resolution full-disk observations, METEOR-Net successfully reconstructs the high-resolution evolution process and is more accurate than other models, nearly matching the ground truth labels. Fig. 10(b) presents a case of temporal super-resolution reconstruction for a highly challenging convective process. In the temporal low-resolution full-disk observations, the extremely low temperature points form on the left side of the convective region at t = T + 15 min. However, in reality, the points form on the right side of the convective region at t = T + 5 min and subsequently move to the left. Except for METEOR-Net, all methods incorrectly identify this, which is related to the linear constraints in these methods. In contrast, METEOR-Net accurately locates the generation position and time of the points by learning the nonlinear dynamics of clouds and water vapor.

To evaluate the temporal super-resolution reconstruction capability of METEOR-Net for convective processes when smaller local regions are used as input, we conduct a set of multiscale input experiments. On the test set, three inference settings are considered: 1) full-patch input, where each 128 × 128 subregion block is super-resolved as a whole; 2) 1/4 local input, where each 128 × 128 block is divided into four 64 × 64 subpatches, which are super-resolved independently and then stitched back to 128 × 128; and 3) 1/16 local input, where each 128 × 128 block is divided into 16 32 × 32 subpatches that are likewise super-resolved independently and then stitched back to 128 × 128.

Quantitative results are reported in Table III. As the input patch size decreases from 128 × 128 to 32 × 32, SSIM only slightly decreases from 0.9948 to 0.9940, while RMSE increases mildly from 0.6816K to 0.7220K. The changes in CSI and POD at different thresholds remain within about 0.01, and FAR increases by only 0.01–0.02. These results indicate that METEOR-Net maintains strong quantitative performance even when inference is performed on smaller local subregions.

All the reconstruction comparison experiments above are conducted under the fixed data split described. To quantify the uncertainty of METEOR-Net with respect to different dataset partitions, a fivefold cross-validation over the years 2018–2022 has further been performed. In each fold, one year is used as the validation set and the remaining four years are used for training, while the test set is still fixed as 2023 and is never involved in training or validation. The mean and standard deviation of each evaluation metric on the test set over the five runs are then computed. As summarized in Table IV, the cross-validation means are very close to the results obtained with the fixed split, and the standard deviations are extremely small, indicating that the performance of METEOR-Net is insensitive to the particular combination of training years and, from another perspective, supporting the use of the fixed split in the main experiments.

F. Ablation Study

An ablation study is conducted to validate the effectiveness of individual components in the METEOR-Net. As shown in Table V, the model architecture is systematically ablated by sequentially removing the MCE mechanism, convective

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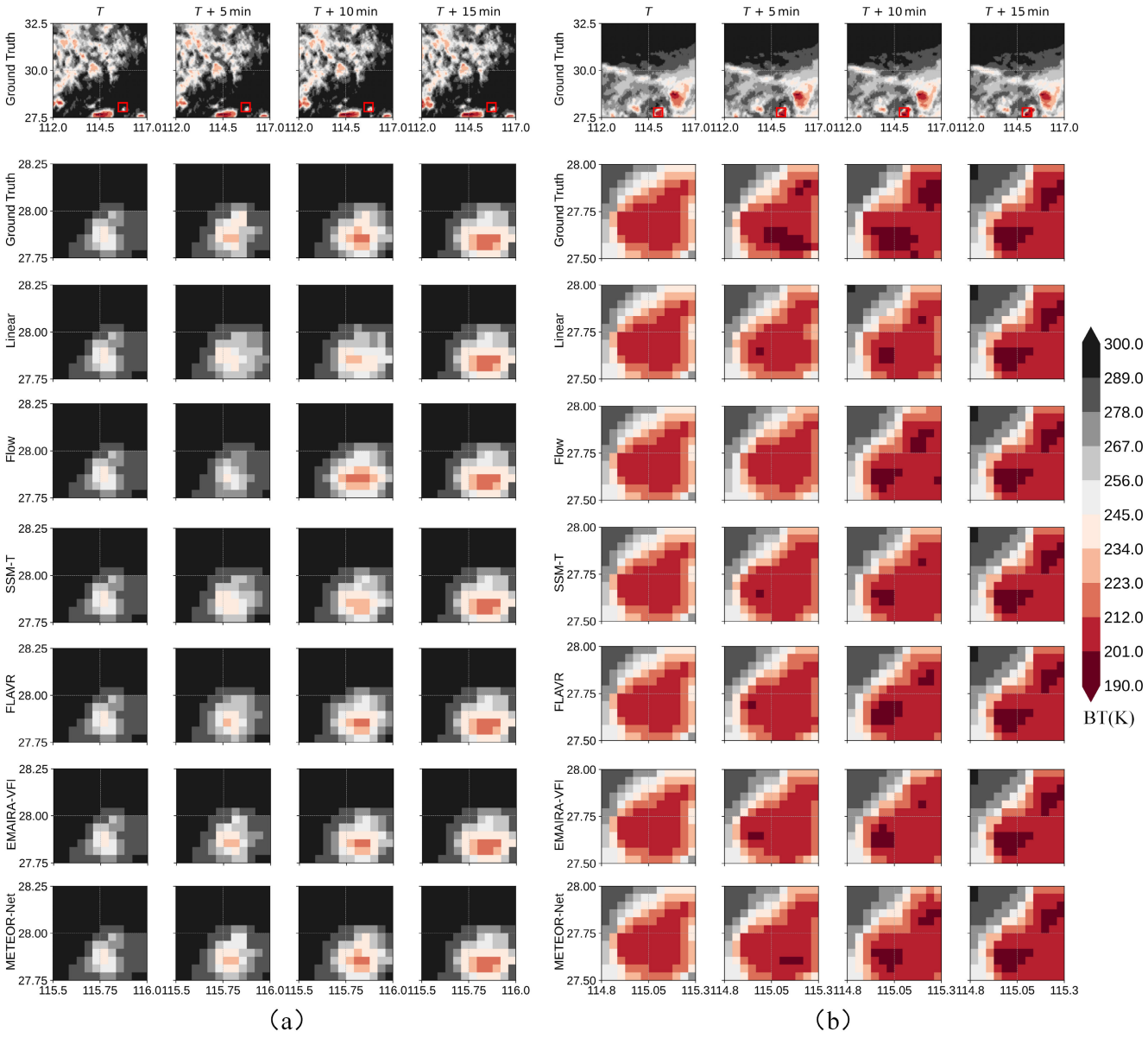


Fig. 10. Case study of temporal super-resolution reconstruction for convective processes. T and T + 15 min represent temporal low-resolution full-disk observations, while T + 5 min and T + 10 min represent ground truth frames or frames reconstructed through temporal super-resolution, and the first row shows the real observation of the convection process, the second row shows an enlarged image of this observation, the subsequent rows display the results of temporal super-resolution reconstruction of the convection process using different methods. (a) Time is from 05:15 UTC to 05:30 UTC on July 24, 2023. (b) Time is from 09:30 UTC to 09:45 UTC on June 30, 2023.

TABLE IV
FIVEFOLD CROSS-VALIDATION STATISTICS OF METEOR-NET

Statistic	SSIM	PSNR	RMSE	CSI			POD			FAR		
				$T<273$	$T<241$	$T<221$	$T<273$	$T<241$	$T<221$	$T<273$	$T<241$	$T<221$
Mean	0.9947	52.11	0.687	0.958	0.876	0.832	0.979	0.930	0.898	0.023	0.070	0.091
Std	0.0001	0.0689	0.0068	0.0006	0.0037	0.0045	0.0014	0.0039	0.0036	0.0020	0.0038	0.0066

region-weighted loss, and temporal-flow evolution mechanism, ultimately reducing to a basic U-shaped encoder–decoder. The progressive performance degradation across all evaluation metrics demonstrates the efficacy of each architectural component and design choice.

G. Quantitative Verification Experiment of Temporal-Flow Evolution

To quantify the benefit of temporal-flow evolution, we conduct a controlled quantitative verification experiment. We construct an optical flow variant, denoted as OF, by replacing

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TABLE V
ABLATION STUDY QUANTITATIVE EVALUATION RESULTS

Module	CSI↑			POD↑			FAR↓		
	T<273	T<241	T<221	T<273	T<241	T<221	T<273	T<241	T<221
-	0.958	0.878	0.835	0.980	0.935	0.900	0.023	0.072	0.089
w/o Mul C	0.951	0.847	0.805	0.975	0.905	0.885	0.027	0.081	0.112
w/o WLoss	0.950	0.827	0.783	0.977	0.883	0.860	0.030	0.091	0.119
w/o ST & Fusion	0.905	0.680	0.667	0.942	0.746	0.736	0.051	0.142	0.130

TABLE VI
FRAME-LEVEL PERFORMANCE UNDER STRONG AND WEAK CONVECTION SUBSETS

Subset	Method	Stat	SSIM ↑	PSNR ↑	RMSE ↓	CSI ↑			POD ↑			FAR ↓		
						$T<273$	$T<241$	$T<221$	$T<273$	$T<241$	$T<221$	$T<273$	$T<241$	$T<221$
Strong Convection	TFEE	Mean	**0.9933**	**49.41**	**0.8922**	**0.9774**	**0.9620**	**0.8473**	**0.9903**	**0.9826**	**0.8997**	**0.0133**	**0.0217**	**0.0750**
		Std	**0.0054**	**2.13**	**0.2865**	**0.0167**	**0.0321**	**0.2090**	**0.0080**	**0.0182**	**0.1957**	**0.0099**	**0.0194**	**0.1335**
	OF	Mean	0.9901	47.75	1.0864	0.9724	0.9502	0.8098	0.9869	0.9755	0.8669	0.0150	0.0274	0.0916
		Std	0.0076	2.34	0.3600	0.0202	0.0417	0.2421	0.0112	0.0253	0.2328	0.0106	0.0237	0.1523
Weak Convection	TFEE	Mean	**0.9931**	**50.68**	**0.7955**	**0.9713**	**0.8302**	-	**0.9872**	**0.9010**	-	**0.0167**	**0.0957**	-
		Std	**0.0083**	**2.79**	**0.3999**	**0.0263**	**0.1949**	-	**0.0136**	**0.1783**	-	**0.0154**	**0.1323**	-
	OF	Mean	0.9896	48.93	0.9812	0.9634	0.7673	-	0.9813	0.8382	-	0.0192	0.1154	-
		Std	0.0126	3.04	0.5006	0.0348	0.2379	-	0.0206	0.2359	-	0.0181	0.1492	-

the temporal-flow evolution module in TFEE with a latent space bidirectional optical flow interpolation module while keeping all other components unchanged. We further define two subsets from the test set, including a strong convection subset with min(T) < 221K and a weak convection subset with 221K ≤ min(T) < 241K. For each subset, we report the mean and standard deviation of frame-level metrics for TFEE and OF. Metrics at T < 221K are not applicable for the weak convection subset by definition. Table VI shows that TFEE consistently achieves higher SSIM and PSNR and lower RMSE on both subsets, and it also performs better on convection process metrics (CSI, POD, and FAR). TFEE further exhibits lower overall standard deviations, which indicates that temporal-flow evolution is more robust and stable than optical flow in both strong and weak convection scenarios.

H. Regional Transfer Generalization Experiments

To evaluate the generalization performance of the model trained over the central China region when applied to temporal super-resolution reconstruction over more complex terrain and different climatic zones, we conduct a set of regional transfer generalization experiments. As shown in Fig. 11, four representative subregions are selected: 1) the Yunnan–Guizhou Plateau (complex high-elevation mountainous terrain); 2) the Yangtze River Delta coastal lowlands (land-sea transition zone); 3) a South China heavy-rainfall region around the Pearl River Delta (frequent severe convection and torrential rainfall); and 4) the semi-arid boundary region along the northern/western margins of the Loess Plateau. These subregions jointly cover typical combinations of terrain and climate from plateau to coastal areas and from humid to semi-arid environments.

For each subregion, we evaluate the temporal super-resolution reconstruction performance of METEOR-Net and

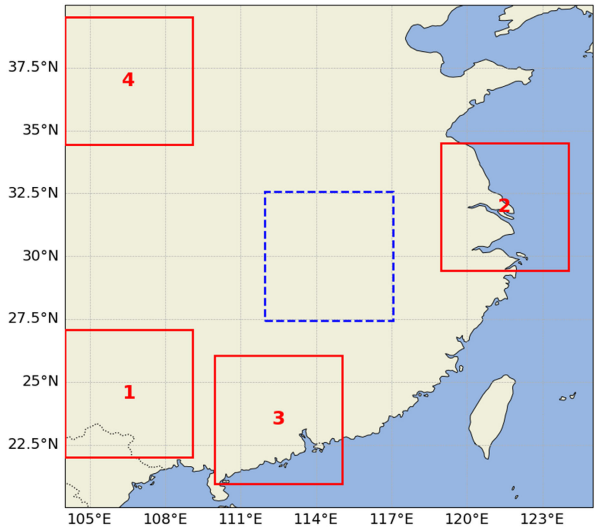


Fig. 11. Schematic locations of subregions. Red boxes denote the subregions used for regional transfer generalization experiments, and the blue dashed box indicates the subregion used for model training.

the comparative methods on the 10.7-μm channel. As summarized in Table VII, METEOR-Net consistently achieves higher SSIM/PSNR and lower RMSE than SSM-T, FLAVR, and EMAIRA-VFI across the four subregions, and yields generally better or comparable CSI/POD/FAR values, indicating robust cross-regional generalization. This indicates that METEOR-Net exhibits good cross-regional generalization ability under markedly different cloud regimes, ranging from plateau and coastal areas to heavy-rainfall and semi-arid regions. At the same time, the absolute metric values differ among subregions, reflecting genuine differences in local cloud evolution characteristics.

To further investigate these regional performance differences and account for the distinct cloud microphysical processes and

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TABLE VII

REGIONAL GENERALIZATION PERFORMANCE OF METEOR-NET AND COMPARATIVE METHODS ON FOUR REPRESENTATIVE SUBREGIONS

Region	Method	SSIM ↑	PSNR ↑	RMSE ↓	CSI ↑			POD ↑			FAR ↓		
					$T<273$	$T<241$	$T<221$	$T<273$	$T<241$	$T<221$	$T<273$	$T<241$	$T<221$
Yunnan–Guizhou Plateau	SSM-T	0.9923	51.03	0.8184	0.9104	0.8476	0.8444	0.9458	0.8884	0.8943	0.0439	0.0622	**0.0699**
	FLAVR	0.9921	51.05	0.8262	0.9109	0.8385	0.8337	0.9451	0.8881	0.8973	0.0430	0.0795	0.0878
	EMAIRA-VFI	0.9929	51.43	0.7790	0.9150	0.8584	0.8513	0.9504	0.9025	0.9046	0.0435	0.0665	0.0732
	Our	**0.9945**	**52.22**	**0.6931**	**0.9258**	**0.8857**	**0.8693**	**0.9624**	**0.9333**	**0.9237**	**0.0428**	**0.0620**	0.0700
Yangtze River Delta Coastal Lowlands	SSM-T	0.9896	49.54	0.9588	0.9396	0.8159	0.7358	0.9640	0.8678	0.7996	0.0308	0.0815	0.1198
	FLAVR	0.9886	49.20	1.0090	0.9373	0.7978	0.7091	0.9622	0.8608	0.7993	0.0317	0.1049	0.1690
	EMAIRA-VFI	0.9919	50.49	0.8459	0.9454	0.8364	0.7537	0.9692	0.8934	0.8291	0.0294	0.0843	0.1306
	Our	**0.9943**	**51.79**	**0.7122**	**0.9560**	**0.8729**	**0.7947**	**0.9767**	**0.9306**	**0.8709**	**0.0241**	**0.0760**	**0.1154**
South China Heavy Rainfall Region	SSM-T	0.9929	50.92	0.7962	0.9125	0.8451	0.8534	0.9467	0.8862	0.9005	0.0431	**0.0646**	0.0653
	FLAVR	0.9925	50.80	0.8188	0.9119	0.8357	0.8448	0.9459	0.8866	0.9054	0.0426	0.0824	0.0813
	EMAIRA-VFI	0.9935	51.35	0.7560	0.9179	0.8575	0.8644	0.9523	0.9038	0.9130	**0.0421**	0.0704	0.0668
	Our	**0.9948**	**52.08**	**0.6839**	**0.9285**	**0.8822**	**0.8775**	**0.9656**	**0.9327**	**0.9273**	0.0432	0.0681	**0.0630**
Semi-Arid Boundary Region	SSM-T	0.9890	49.03	1.0087	0.9429	0.8264	0.6310	0.9655	0.8821	0.7128	0.0282	0.0828	0.1687
	FLAVR	0.9882	48.83	1.0457	0.9421	0.8145	0.6048	0.9654	0.8775	0.7126	0.0295	0.0947	0.2298
	EMAIRA-VFI	0.9902	49.54	0.9486	0.9459	0.8371	0.6472	0.9688	0.8954	0.7439	0.0285	0.0818	0.1855
	Our	**0.9917**	**49.91**	**0.9003**	**0.9501**	**0.8519**	**0.6797**	**0.9718**	**0.9127**	**0.7749**	**0.0267**	**0.0816**	**0.1650**

TABLE VIII

PERFORMANCE OF METEOR-NET BEFORE AND AFTER SMALL-SAMPLE REGIONAL FINE-TUNING ON FOUR REPRESENTATIVE SUBREGIONS

Region	Setting	SSIM ↑	PSNR ↑	RMSE ↓	CSI ↑			POD ↑			FAR ↓		
					$T<273$	$T<241$	$T<221$	$T<273$	$T<241$	$T<221$	$T<273$	$T<241$	$T<221$
Yunnan-Guizhou Plateau	Base	0.9945	52.22	0.6931	0.9258	0.8857	0.8693	**0.9624**	0.9333	0.9237	0.0428	0.0620	**0.0700**
	Fine-tuned	**0.9950**	**52.53**	**0.6631**	**0.9290**	**0.8913**	**0.8797**	0.9606	**0.9390**	**0.9445**	**0.0373**	**0.0608**	0.0766
Yangtze River Delta Coastal Lowlands	Base	0.9943	51.79	0.7122	0.9560	0.8729	0.7947	0.9767	**0.9306**	**0.8709**	0.0241	0.0760	0.1154
	Fine-tuned	**0.9949**	**52.13**	**0.6778**	**0.9581**	**0.8803**	**0.8116**	**0.9778**	0.9221	0.8599	**0.0222**	**0.0569**	**0.0782**
South China Heavy Rainfall Region	Base	0.9948	52.08	0.6839	0.9285	0.8822	0.8775	0.9656	0.9327	0.9273	0.0432	0.0681	**0.0630**
	Fine-tuned	**0.9951**	**52.30**	**0.6618**	**0.9295**	**0.8873**	**0.8853**	**0.9656**	**0.9355**	**0.9426**	**0.0417**	**0.0636**	0.0689
Semi-Arid Boundary Region	Base	0.9917	49.91	0.9003	0.9501	0.8519	0.6797	0.9718	0.9127	0.7749	0.0267	0.0816	0.1650
	Fine-tuned	**0.9945**	**51.63**	**0.7147**	**0.9581**	**0.8867**	**0.7358**	**0.9742**	**0.9304**	**0.8304**	**0.0193**	**0.0549**	**0.1429**

triggering mechanisms in different regions, we also examine the effect of small-sample regional fine-tuning. For each subregion, we randomly sample about 10% of the available training sequences and perform a lightweight fine-tuning of the METEOR-Net model originally trained over central China, using only about 5% of the total training iterations. As reported in Table VIII, SSIM, PSNR, and RMSE are clearly improved in all four subregions after fine-tuning, and CSI scores increase overall; POD and FAR also move toward more favorable values at most thresholds. These results show that, starting from a unified model, low-cost regional fine-tuning with a small amount of local data can effectively adapt METEOR-Net to convective cloud evolution under different terrain and climate backgrounds, and significantly mitigate performance gaps among different cloud regimes. Taken together, these additional experiments provide a more comprehensive assessment of the model's regional transfer generalization ability over complex terrain and diverse climate zones, and offer practical guidance for further optimizing the model in specific key regions with limited local samples.

I. Independent Validation With Radar and Precipitation-Based Tasks

To further assess the physical credibility of the reconstructed fields, we introduce external datasets, including Doppler weather radar and surface precipitation observations from national meteorological stations, to independently validate the temporal super-resolution results.

As illustrated in Fig. 12, two representative severe convective cases are selected. In each case, the background shows the FY-4A 10.7-μm brightness temperature field, and the green mask denotes composite radar reflectivity exceeding 35 dBZ. Time instants at multiples of 15 min correspond to the original FY-4A observations, while the intermediate time steps are the temporally super-resolved reconstructions produced by METEOR-Net. In both examples, strong radar echoes exhibit good spatial correspondence with regions of markedly low brightness temperature, and this relationship is maintained consistently over time. Moreover, no large areas of spuriously cold cloud tops are found in regions where no pronounced radar echoes are present. These results indicate that, within the AGRI observation framework, METEOR-Net provides a physically reasonable description of the evolution of deep convective cloud systems, rather than learning spurious patterns that contradict the physical laws of atmospheric convection.

To further validate the physical realism of the temporally super-resolved reconstructions from a more quantitative perspective, we perform an additional independent evaluation based on surface precipitation observations from national meteorological stations. Specifically, two downstream tasks are designed: station-based severe convective weather

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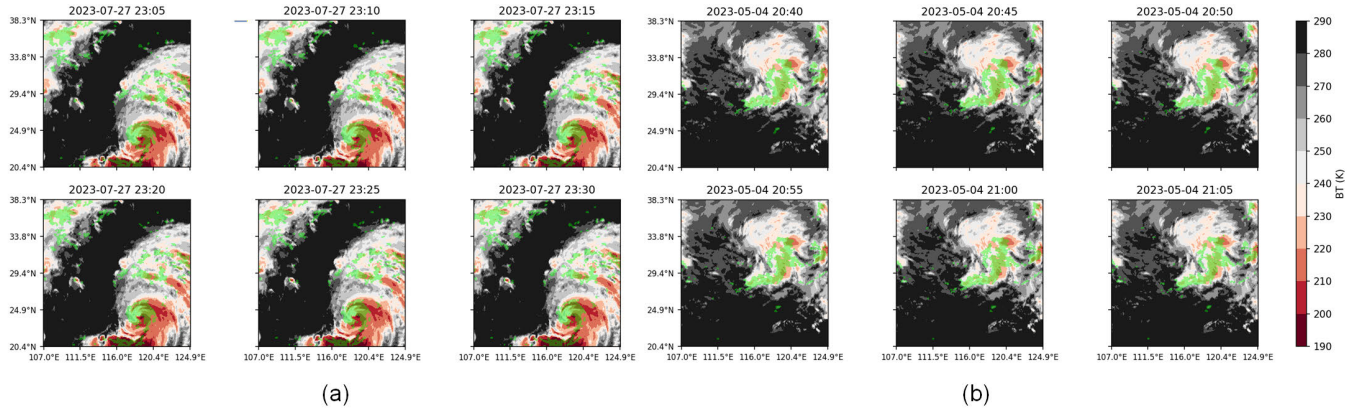


Fig. 12. Two representative severe convective cases. Background shows the FY-4A/AGRI 10.7-μm brightness temperature field, and the green mask denotes composite radar reflectivity ≥35 dBZ. (a) Typhoon rainband case from 23:05 to 23:30 UTC on July 27, 2023. (b) Mesoscale convective system case from 20:40 to 21:05 UTC on May 4, 2023.

identification and station-based severe convective weather prediction. The experimental domain covers eastern and central China (20.40°–38.28°N, 107.04°–124.92°E) and contains nearly 1000 national meteorological stations. The dataset is split chronologically, with data from 2018 to 2021 used for training, 2022 for validation, and 2023 for testing. For both tasks, the objective is to determine, for all stations within the domain, whether severe convective weather is occurring at the current time or will occur within the next 1–3 h. Severe convective weather is defined as the occurrence of heavy precipitation ≥ 20 mm/h within a 40-km radius around a station.

In both downstream tasks, we adopt the same U-Net architecture and training configuration, and only vary the input sequences during training/validation/testing among three options: 1) the original FY-4A sequences with 15-min temporal resolution (raw); 2) 5-min sequences obtained by simple temporal linear interpolation (linear); and 3) 5-min temporally super-resolved sequences produced by METEOR-Net (our). Since severe convective weather identification and prediction must extract convective signals that are closely related to the occurrence of heavy surface precipitation from the temporal evolution of satellite cloud fields, performance differences under the same network architecture and training setup mainly reflect differences in the completeness of physically relevant information and the noise level carried by the respective input sequences. Therefore, superior performance in these downstream tasks indicates that the input sequence provides a more realistic and detailed description of convective development and is more physically consistent with the independent surface precipitation observations.

Table IX compares the results of the severe convective weather identification task. Because temporal linear interpolation is only a simple weighted average of neighboring frames, it introduces erroneous convective structures and degrades the quality of the convective signal. As a result, using the linearly interpolated sequences as input leads to lower CSI and POD and higher FAR than using the original 15-min sequences. In contrast, when the METEOR-Net super-resolved sequences are used as input, CSI and POD are improved, and FAR is reduced relative to the raw input. This indicates that the convective signals reconstructed by METEOR-Net are more consistent with the actual occurrence of heavy precipitation and provide more detailed, physically meaningful information for severe convective weather identification that cannot be fully captured from the original 15-min sequences alone.

TABLE IX
SEVERE CONVECTION IDENTIFICATION PERFORMANCE WITH DIFFERENT INPUT SEQUENCES

Method	CSI ↑	POD ↑	FAR ↓
Raw	0.471	0.660	0.378
Linear	0.466	0.654	0.382
Our	**0.479**	**0.668**	**0.371**

Fig. 13 presents the comparison results for the severe convective weather prediction task. As expected, CSI and POD decrease with increasing lead time for all methods, while FAR increases. The METEOR-Net input (our) consistently achieves the highest CSI at lead times of 1, 2, and 3 h, and achieves the highest POD at 2 and 3 h, while maintaining the lowest FAR at all three lead times. These findings demonstrate that, under the independent constraint of surface precipitation observations, the temporally super-resolved sequences produced by METEOR-Net provide substantial benefits for short-term severe convection forecasting, rather than merely yielding visually smoother images or fitting noise.

J. Channel Collaboration Interpretability Analysis

As shown in Fig. 14, heat map analysis reveals several interesting findings, particularly in channel 12, a primary infrared imaging channel. During channel 12 reconstruction, a notable weight disparity emerges between channels 9 and 10. Channel 12 (centered at 10.7 μm) operates within the infrared window region with minimal atmospheric absorption, enabling effective cloud-top temperature measurement and cloud detection. In contrast, Channel 9 (6.25 μm) primarily detects upper-level water vapor, influencing large-scale atmospheric patterns while exhibiting limited direct impact on local

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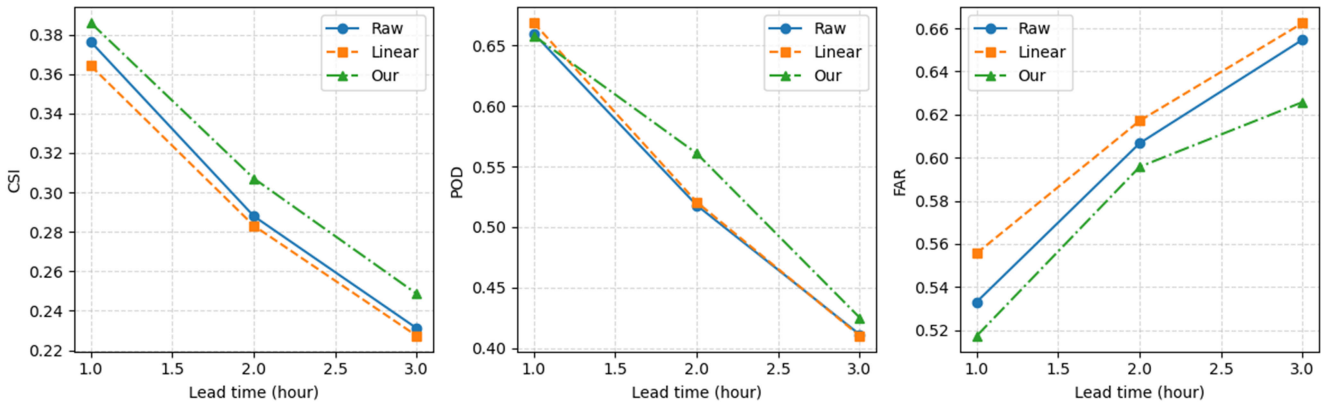


Fig. 13. Quantitative comparison of the 1–3 h severe convective weather prediction task. Raw denotes using the original FY-4A sequences with 15-min temporal resolution as input, Linear denotes using 5-min sequences obtained by temporal linear interpolation as input, and Our denotes using 5-min sequences generated by METEOR-Net as input.

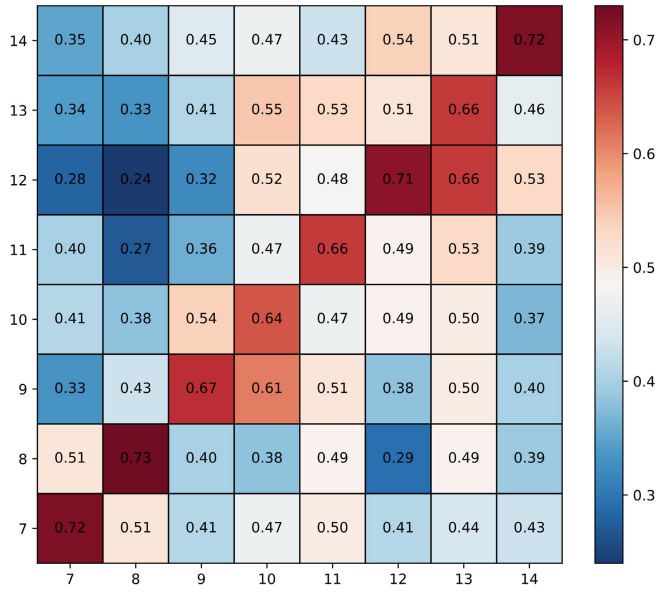


Fig. 14. Collaborative channel contribution heat maps during temporal super-resolution reconstruction. The contribution parameters are derived from adaptive channel weighting values in METEOR-Net's MCE module. Each row represents a specific channel's reconstruction process, with columns indicating the collaborative contribution degrees of all channels to that particular reconstruction process.

cloud formations. Channel 10 (7.1 μm), focused on mid-level water vapor detection, directly influences atmospheric layer stability through moisture content. Enhanced mid-level humidity provides increased water vapor supply, facilitating latent heat release through vapor condensation and strengthening convective intensity. Despite their proximate wavelengths within the water vapor channel, channel 10 demonstrates higher auxiliary significance than channel 9 in channel 12 reconstruction, aligning with the model's adaptive weighting mechanism.

In addition, channel 13, adjacent to channel 12 and traditionally employed for water vapor correction in cloud detection, exhibits elevated correlation with water vapor channels due to enhanced absorption sensitivity. This correlation manifests in the channel weights during water vapor channel reconstruction. These findings suggest that METEOR-Net may have captured the underlying mechanisms of convective cloud formation and development during its channel collaborative fusion and temporal-flow evolution processes for temporal super-resolution reconstruction.

V. Conclusion and Future Work

In this article, a temporal super-resolution framework for geostationary satellite data based on spatiotemporal self-supervised learning is proposed, addressing limitations in conventional VFI frameworks through comprehensive improvements in data construction, model architecture, and training-evaluation strategy. In addition, a novel model, METEOR-Net, is introduced for temporal super-resolution of geostationary satellite cloud data. METEOR-Net is a flow-based temporal super-resolution method that replaces optical flow-based frame interpolation by estimating and evolving temporal-flows, thereby better adapting to the data characteristics of geostationary satellites. Within this framework, extensive quantitative and qualitative experiments indicate that METEOR-Net yields temporal super-resolution reconstruction results that surpass existing temporal super-resolution methods and models for geostationary satellites in terms of visual fidelity and the accuracy of convective process reconstruction. Regional transfer generalization tests, together with additional experiments over subregions with different terrain and climatic conditions and small-sample regional fine-tuning, provide a comprehensive assessment of the model's regional generalization ability and indicate good potential for regional transferability. Although an exhaustive evaluation across all possible climate zones has not yet been conducted, these regional experiments suggest a practically feasible usage pattern: first, pretrain a general model over a broad domain, and then perform lightweight fine-tuning with a small number of local samples in the target region, thereby striking a balance between computational cost and regional adaptability.

In addition, an interpretability analysis of METEOR-Net's temporal super-resolution reconstruction process is conducted.

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At the same time, the physical credibility of the temporally super-resolved fields produced by METEOR-Net is further supported by independent validations using radar and precipitation-based tasks.

This study improves the temporal resolution of FY-4A cloud data from 15 to 5 min, which alleviates the temporal sparsity of geostationary measurements and provides higher temporal-resolution cloud observation sequences for future convective weather identification and warning. Case analyses of convective evolution show that METEOR-Net reconstructs the cloud-top cooling evolution associated with early deep convection formation within the 15 min observation gap. Under more challenging rapidly developing conditions, it also captures the formation location and timing of the very low brightness temperature core and its subsequent displacement.

Independent validation based on radar reflectivity consistency checks and station-precipitation-based downstream identification and prediction tasks further indicates that the reconstructed fields exhibit good spatiotemporal correspondence with regions of intense radar echoes, do not produce widespread spurious cold cloud tops, and provide more physically meaningful information for downstream applications.

Although the temporal super-resolution model performs well in reconstruction experiments, we find that its reconstruction errors remain nonnegligible for operational applications. The multichannel error maps in the experiments show that large errors tend to concentrate in regions with rapid evolution. This challenge is most pronounced during convective initiation and rapid development, when cloud systems exhibit strong nonlinearity and high temporal variability, which makes intermediate frame reconstruction more difficult. During initiation, the cold cloud-top signal can be weak and confined to a small fraction of pixels, which increases sensitivity to small perturbations. During rapid development, fast nonlinear evolution, and displacement can increase the risk of spatiotemporal biases.

Based on these characteristics, we recommend combining the 5-min super-resolved frames with the original FY-4A observations at the 15-min anchor times to perform temporal-consistency checks and quality control, with particular attention to the propagation of reconstruction errors into downstream applications. For example, convective identification can suffer from missed detections or false alarms due to biases in cold cloud-top location or intensity, and short-term precipitation nowcasting can also be affected. For critical operational applications, it is further advisable to incorporate uncertainty-aware safeguards, including multisource information fusion such as radar and numerical weather prediction (NWP), multimodel or ensemble consistency checks, and confidence-based quality control, to flag and manage low-confidence reconstructions.

Finally, in this study, METEOR-Net is instantiated to perform temporal super-resolution from 15 to 5 min within each full-disk revisit interval. Specifically, two intermediate frames at T + 5 min and T + 10 min are reconstructed using 5-min FY-4A/AGRI REGC observations as temporal high-resolution ground truth labels. Nevertheless, the sequence-flow-based architecture itself is not intrinsically restricted to a fixed temporal step and can, in principle, be extended via time conditioning to interpolate intermediate frames at arbitrary time points within the revisit window. At present, this capability cannot be rigorously validated on FY-4A because the available ground truth observations have a temporal resolution of only 5 min. The next-generation geostationary satellite FY-4B, however, can provide regional observations with a temporal resolution of 1 min. In future work, we plan to migrate the proposed temporal super-resolution framework to FY-4B data and to conduct more stringent observational validation of intermediate reconstructions at finer time resolutions, further assessing the physical fidelity of METEOR-Net's arbitrary time interpolation for rapidly evolving convective processes.

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